

ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

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Assignment on **Fuzzy Logic and Reasoning**

Q2a. Solution:

Rule 1 - Consequent Membership Function:

$$\begin{aligned} B1 &= \text{TrapMF}(y, [0, 2, 4, 6]) \\ \text{Centroid: } &(6 - 0) / 2 = 3 \\ \text{Area: } &[(2 + 6) / 2] * 1 = 4 \end{aligned}$$

Rule 2 - Consequent Membership Function:

$$\begin{aligned} B2 &= \text{TriMF}(y, [4, 6, 8]) \\ \text{Centroid: } &(4 + 8) / 2 = 6 \\ \text{Area: } &(8 - 4) / 2 = 2 \end{aligned}$$

Rule 3 - Consequent Membership Function:

$$\begin{aligned} B3 &= \text{TrapMF}(y, [6, 8, 9.6, 11.6]) \\ \text{Centroid: } &(6 + 11.6) / 2 = 8.8 \\ \text{Area: } &[((11.6 - 6) + (9.6 - 8)) / 2] * 1 = 3.6 \end{aligned}$$

$$\text{Sum of the three centroids of areas: } 3 + 6 + 8.8 = \mathbf{17.8000}$$